MATHEUS GOMES FÉLIX DA SILVA

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Languages Portuguese (native), English (upper-intermediate).
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SUMMARY

Computer Science student at UFPB and volunteer Machine Learning Engineer at the ARIA lab, focused on computer vision (detection, segmentation and OCR) and predictive modeling. Full-stack experience as co-founder and developer of Compendium. Communicative, disciplined and driven to build tech ventures.

EDUCATION

Federal University of Paraíba (UFPB) — B.Sc. in Computer Science Nov 2025 – Jun 2029 (expected)

EXPERIENCE

Machine Learning Engineer (Volunteer) - ARIA Lab [Remote] May 2026 – Present

· Develop computer vision models that read drug names and dosages from images of syringes, infusion pumps, IV bags, vials and pens, in partnership with Hospital Israelita Albert Einstein.
· Train and compare object detection and segmentation models (YOLO, RF-DETR) to identify objects and liquids in hospital settings.
· Integrate OCR into the detection pipeline and run benchmark analyses to compare architectures and guide model selection.
· Build models to analyze and forecast financial data, such as bank customer trust scores and expected credit limits.

Co-founder & Developer - Compendium Mar 2026 – Sep 2026

· Co-founded and developed a multi-tenant web application (one subdomain per client) that centralizes policies, procedures, meetings, warnings and suspensions.
· Implemented policy read-tracking through legally valid electronic signatures (Autentique), with signed PDFs stored on Cloudflare R2.
· Built role and permission management, defining what each role can view and edit in the system.
· Integrated billing (Asaas), email delivery (Resend) and PWA push notifications on top of a REST API with JWT authentication.
· **Stack:** JavaScript, Node.js (Fastify), React (Vite, Tailwind CSS), PostgreSQL, Prisma, Railway.

PROJECTS

Financial Manager Python, Rich

· Financial management system for small businesses: income, expenses, balance, reports and editable categories.
· Modular architecture of about 3,000 lines, with input validation, JSON persistence and test data generation.

Customer Manager Python, Rich

· CRUD customer registry with validation of CPF (Brazilian tax ID), phone and email, and duplicate prevention.

SKILLS

Programming	Python, JavaScript, Java, SQL.
Machine Learning	Computer vision, detection and segmentation, OCR, YOLO, RF-DETR, benchmarking, predictive modeling.
Web development	Node.js, Fastify, React, Tailwind CSS, REST APIs, PostgreSQL, MySQL, Prisma.
Automation	n8n (beginner): building workflows and integrating services.
Soft skills	Communication, interpersonal relationships, discipline and self-management.